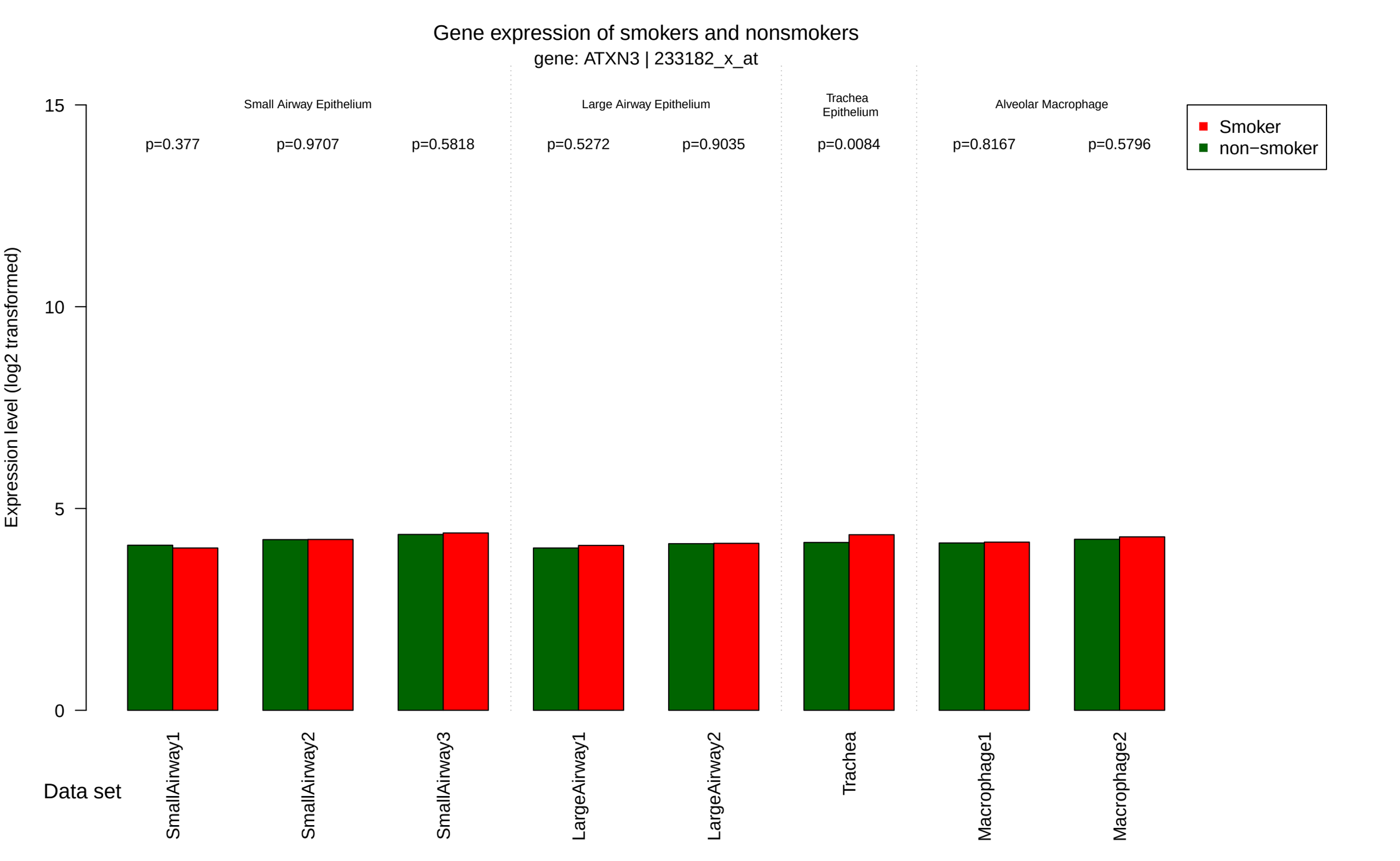
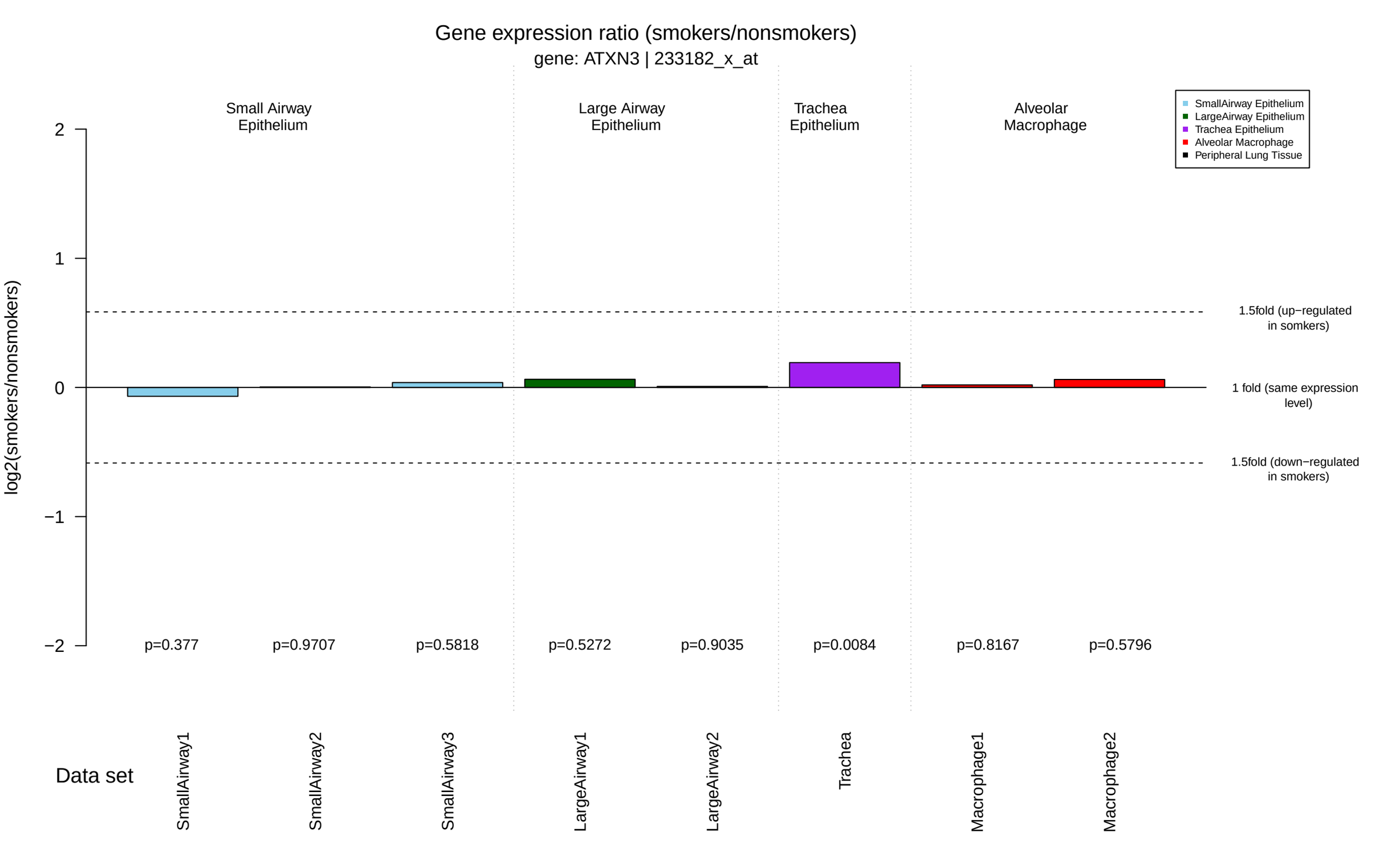
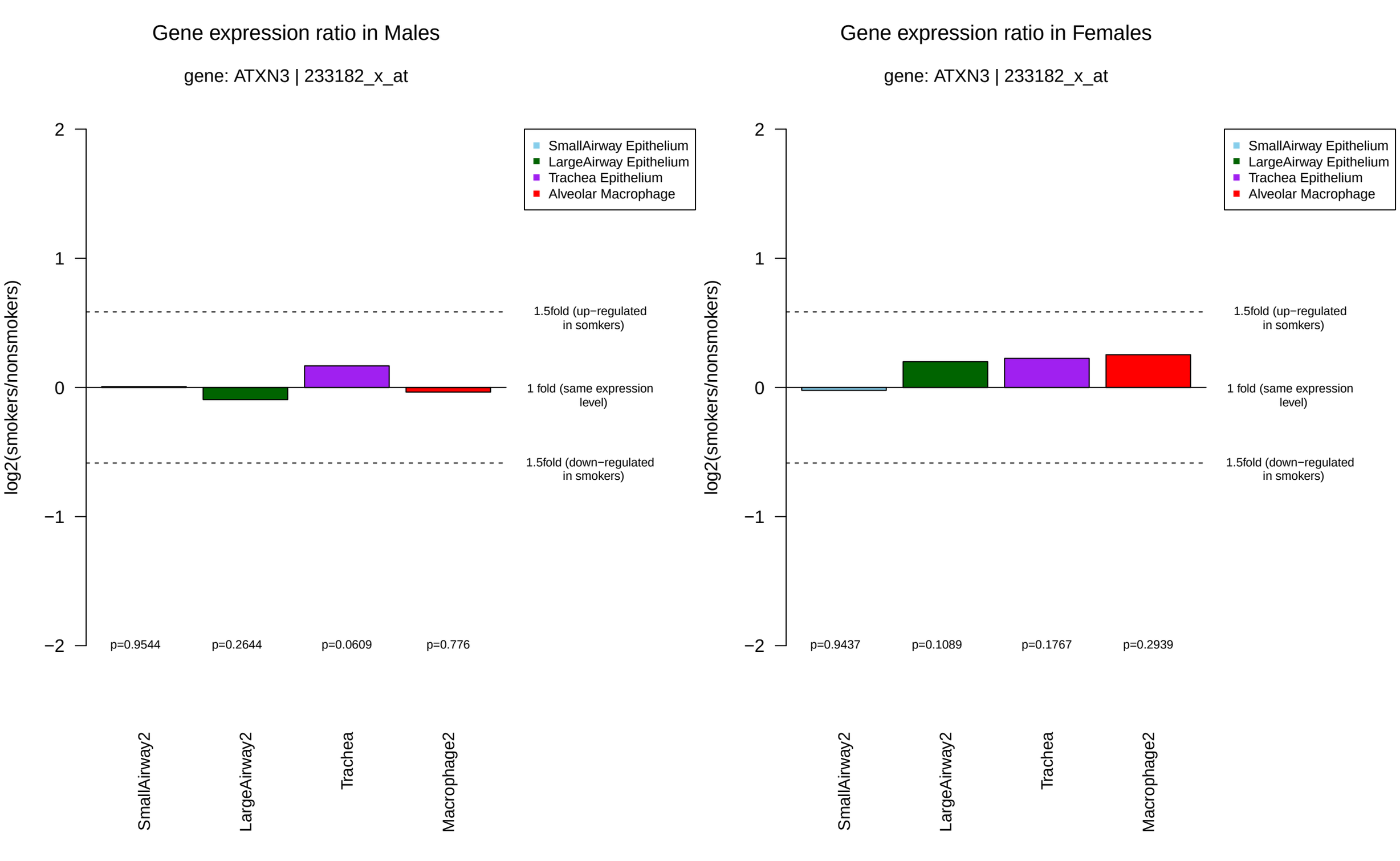
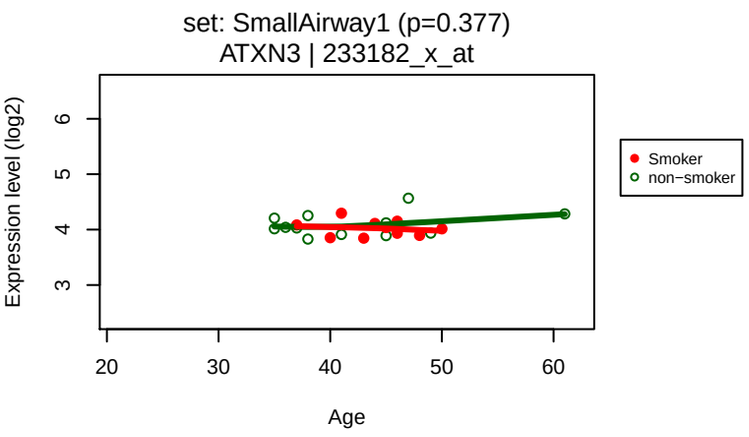
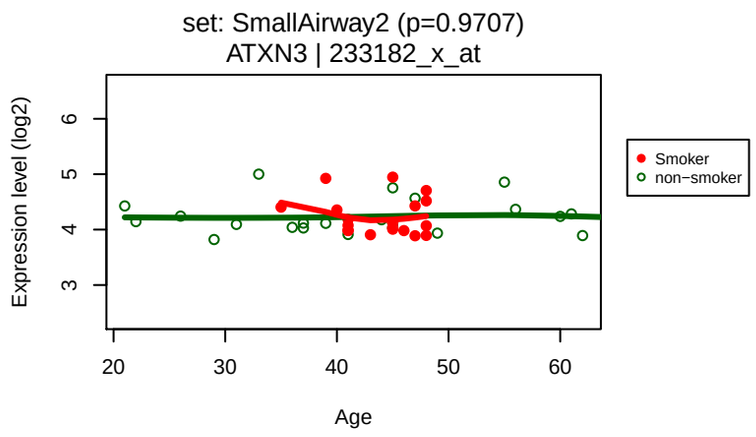
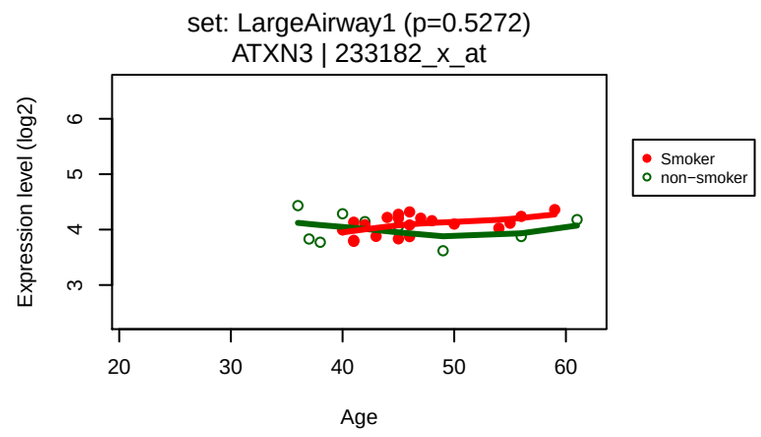

Gene expression of smokers and nonsmokers
gene: ATXN3 | 233182_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.377
p=0.9707
p=0.5818
p=0.5272
p=0.9035
p=0.0084
p=0.8167
p=0.5796
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

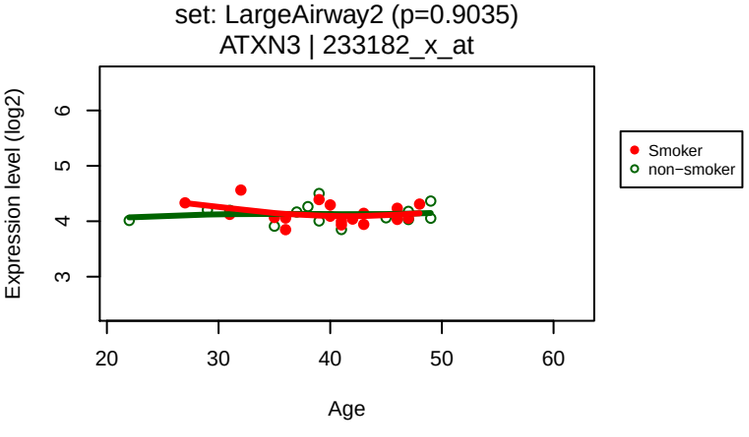

Gene expression ratio (smokers/nonsmokers)
gene: ATXN3 | 233182_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.377
p=0.9707
p=0.5818
p=0.5272
p=0.9035
p=0.0084
p=0.8167
p=0.5796
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

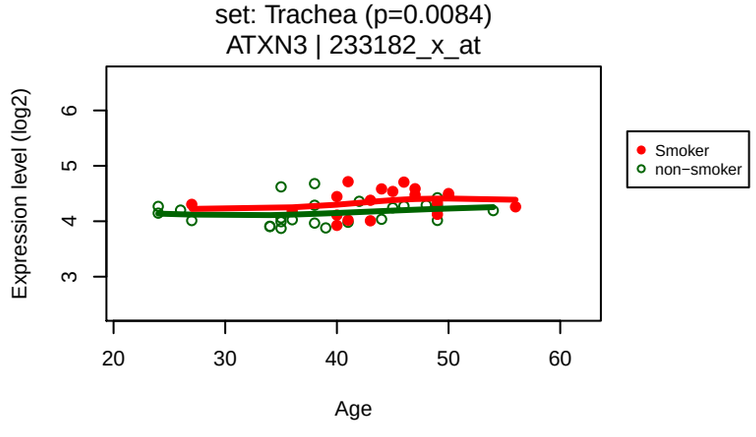

Gene expression ratio in Males
gene: ATXN3 | 233182_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9544
p=0.2644
p=0.0609
p=0.776
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATXN3 | 233182_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9437
p=0.1089
p=0.1767
p=0.2939
SmallAirway2
LargeAirway2
Trachea
Macrophage2

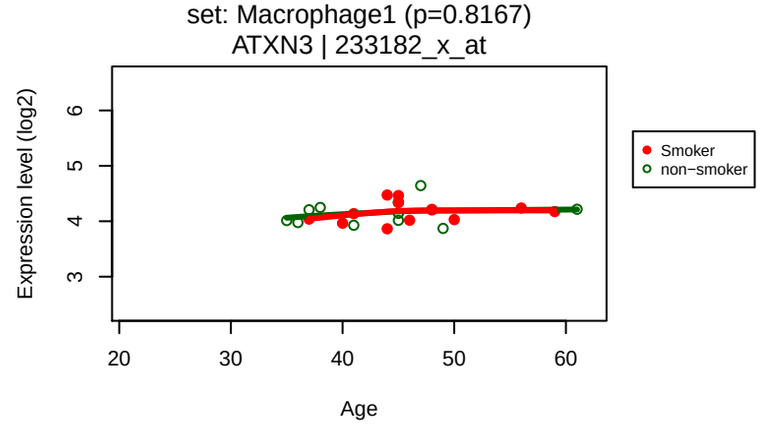

set: SmallAirway1 (p=0.377)
ATXN3 | 233182_x_at
Expression level (log2)
Age
Smoker
non−smoker

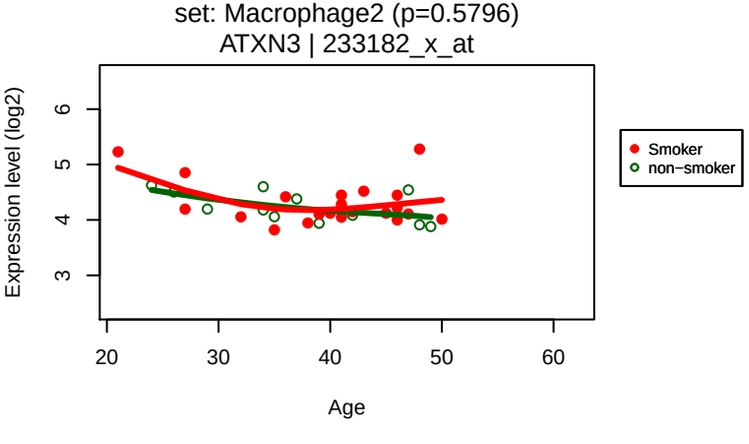

set: SmallAirway2 (p=0.9707)
ATXN3 | 233182_x_at
Expression level (log2)
Age
Smoker
non−smoker

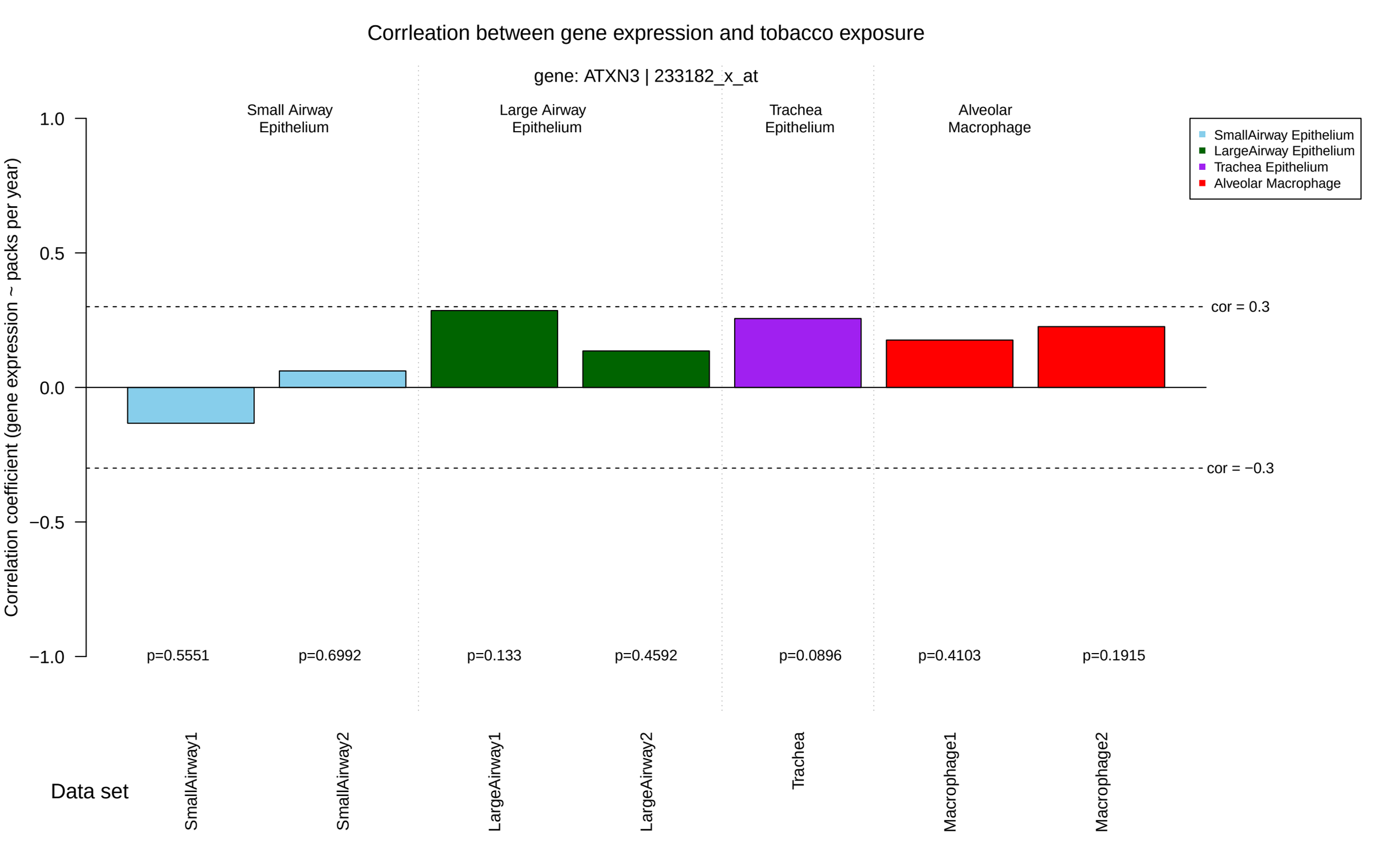

set: LargeAirway1 (p=0.5272)
ATXN3 | 233182_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9035)
ATXN3 | 233182_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0084)
ATXN3 | 233182_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8167)
ATXN3 | 233182_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5796)
ATXN3 | 233182_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATXN3 | 233182_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5551
p=0.6992
p=0.133
p=0.4592
p=0.0896
p=0.4103
p=0.1915
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2